

CENTRALIZED TRAINING AND DECENTRALIZED EXECUTION OF MARL AGENT USING VDN ARCHITECTURE

Group 25-26J-473

Project Final Thesis

Fernandopulle S N -IT22000576

B.Sc. (Hons) in Information Technology Specializing in
Software Engineering

Department of Computer Science & Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

March 2026

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1. Declaration page of the candidates & supervisor

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The above candidate has carried out this research thesis for the Degree of Bachelor of Science (honors) Information Technology (Specializing in Software Engineering) under my supervision.

.....

Signature of the supervisor:

Date.....

(Prof Nuwan Kodagoda)

.....

Signature of the Co-supervisor:

Date.....

(Ms. Karthiga Rajendran)

2. Abstract

Traffic congestion is a major challenge in urban areas in Sri Lanka, where traditional fixed-time traffic signal systems fail to adapt to dynamic conditions such as pedestrian density at pedestrian crossings, emergency vehicle priority systems, and accident management. This report presents a cooperative multi-agent reinforcement learning system employing centralized training with decentralized execution and value decomposition networks to interconnected junctions. Each intersection operates as an independent agent, learning adaptive signal policies through a priority-weighted reward function that simultaneously optimizes emergency vehicle passage, pedestrian safety across vulnerability categories, and overall network throughput. The system integrates four real-time computer vision-based detection streams of emergency vehicles, normal vehicles, pedestrians, and accidents via a continuous database injection pipeline into a stimulatory environment called SUMO.

A lightweight inter-agent communication protocol enables proactive coordination for emergency vehicle green waves without centralized computation during execution. A production-ready framework incorporating operational mode switching, real-time WebSocket dashboard visualization for traffic authorities, and IoT synchronization for physical signal displays supports gradual deployment alongside existing infrastructure.

Keywords - Traffic Congestion, Multi-Agent Reinforcement Learning (MARL), SUMO Simulation, Value Decomposition Architecture (VDN), Centralized training and decentralized execution (CTDE)

3. Acknowledgement

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4. List Of Abbreviations

SUMO	Simulation of Urban Mobility
TraCI	Traffic Control Interface
MARL	Multi-Agent Reinforcement Learning
ML	Machine Learning
NN	Neural Network
CTDE	Centralized Training and Decentralized Execution
VDN	Value decomposition architecture
DL	Deep Learning

Table 1 : List of Abbreviations

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1. Introduction

7.1. Background Study and Literature Review

7.1.1. Background Study

Urban traffic congestion has become an increasingly critical concern in Sri Lanka, particularly in rapidly developing suburban corridors surrounding Colombo. The current traffic signal control infrastructure across the island relies predominantly on fixed-time control systems, where each junction operates on a predetermined cycle that allocates equal green time to all approaching lanes regardless of actual demand, that vary based on time-of-day schedules, typically shorter cycles during night hours, standard timing during daytime, and extended phases during peak congestion periods [1]. Pressure is increasing due to rapid urbanization and rising vehicle ownership. The vehicle composition is highly heterogeneous, comprising private cars, public buses, school vans, three-wheelers, motorcycles, trucks, and lorries, each with distinct acceleration profiles, space occupancy, and priority requirements, which is being disregarded by the current traffic control system.

Additionally, pedestrian activity is substantial, particularly near educational institutions, commercial areas, and public transport hubs, yet pedestrian crossing phases are often absent or inadequately timed. In situations of extreme congestion, vehicle breakdowns, or accidents, traffic management defaults to manual control by police officers who direct vehicles using hand signals, which is a practice that, while providing human adaptability, introduces its own inefficiencies due to the inability of a single officer to coordinate across multiple interconnected junctions simultaneously. Currently, this traditional traffic control system is increasingly being upgraded to automated, sensor-based systems.

7.1.2. Literature Review

Recent studies have explored the application of intelligent traffic signal control using machine learning and computer vision techniques. In [7], a YOLO-based real-time adaptive traffic signal system is proposed for a single intersection in Sri Lanka. The system utilizes CCTV-based vehicle detection to dynamically adjust green phase

durations according to real-time demand, demonstrating significant reductions in queue delay and travel time under simulated conditions. However, the approach is limited to an isolated intersection and does not consider coordination with neighboring junctions or broader network-level optimization. Similarly, [8] investigates the use of single-agent deep Q-learning for traffic signal control in a multi-intersection environment. Each intersection is modeled as an independent agent that learns optimal policies using a standard deviation-based reward function. While improvements in fuel consumption and travel time are reported, the lack of inter-agent communication means that each agent treats others as part of the environment, which may introduce non-stationarity and limit coordinated decision-making across the network.

To address coordination, [9] proposes a VDN-based MARL framework for two interconnected intersections. Their approach incorporates multiple reward components, including delay minimization and progression incentives, and extends functionality to include transit signal priority. Although the framework demonstrates performance improvements, the reliance on manually engineered reward structures tailored to specific road configurations may affect scalability and adaptability to different traffic conditions. Further advancing multi-agent coordination, [10] introduce QRC-TSC, a communication-aware MARL framework built on a centralized QMIX architecture. The model enables agents to learn when and what information to communicate, improving communication efficiency while maintaining low congestion levels. Despite these advantages, the framework primarily focuses on communication optimization and employs a simplified reward structure based on queue length, which does not capture diverse traffic priorities such as emergency vehicles or pedestrian flows.

7.2. Research Gap

Sensory-based intersection control, Sensory-based vehicles and driver identification and control, single-agent reinforcement learning junction control are the focus of current traffic optimization techniques as ongoing research islandwide. International research includes the scope of MARL Based on Representational Communication for Large-Scale

Traffic Signal Control. They lack integrated emergency and pedestrian safety management, real-time multimodal perception, and coordinated multi-intersection control. Additionally, priority-based traffic management in which high-occupancy vehicles and public transportation are given precedence depending on passenger capacity is not taken into account by current methods. No prior work has applied MARL-based traffic control to Sri Lankan traffic conditions, which differ significantly from the North American and European contexts where most research has been conducted, particularly in terms of vehicle composition heterogeneity, pedestrian density, and traffic enforcement patterns.

Method	Strengths	Research Gaps	References
YOLO-based real-time adaptive traffic signal control using CCTV vehicle detection for a single intersection	-Real-time vehicle detection using computer vision -Dynamically adjusts green signal durations -Reduces queue delay and travel time under simulation	✗ Limited to a single intersection ✗ No coordination with neighboring junctions ✗ Lacks network-level optimization	[5] [6]
Single-agent Deep Q-Learning (DQN) for multi-intersection traffic control (independent agents)	-Applies reinforcement learning to multiple intersections -Improves fuel consumption and travel time -Simple and scalable per intersection	✗ No inter-agent communication ✗ Treats other agents as environment (non-stationarity issue) ✗ Limited coordination across intersections	[7] [8]
VDN-based Multi-Agent Reinforcement	-Enables coordination between agents	✗ Reward functions are manually engineered ✗ Limited scalability to	[9] [10]

Learning (MARL) for two interconnected intersections	-Uses multiple reward components (delay, progression) -Supports transit signal priority	diverse road networks ✗ Tested on only two intersections	
QRC-TSC: Communication-aware MARL using QMIX architecture	- Learns efficient communication between agents - Reduces congestion effectively - Optimizes what and when to communicate.	✗ Focuses mainly on communication efficiency ✗ Uses simplified reward (queue length only) ✗ Does not consider emergency vehicles or pedestrians	[11] [12] [13]

Table 2 : Competitive Analysis

7.3. Research Problem

This traditional approach to traffic signal control creates several interconnected problems that compound across the urban road network. Fixed-time signal plans operate on static schedules and cannot respond to real-time fluctuations in vehicle demand. Whether an approach has zero vehicles or is heavily congested, the allocated green time remains unchanged, resulting in wasted green time on empty approaches while queued vehicles on other approaches accumulate delay. During unexpected events such as vehicle breakdowns, accidents, or sudden demand surges from nearby events, the system offers no mechanism for adaptation. Each traffic light in the current system operates independently, without any observation of or communication with neighboring junctions. A signal decision at one intersection, such as releasing a large platoon of vehicles, directly affects the downstream junction within seconds, yet the downstream controller has no advance knowledge and cannot prepare its phase timing accordingly. This lack of coordination creates cascading congestion waves where blockages at one intersection propagate to adjacent junctions, ultimately contributing to city-wide congestion rather than localized delays.

The fixed-time system treats all vehicles equally, providing no mechanism to prioritize emergency vehicles, public transit, or vulnerable road users. An ambulance carrying a critical patient receives the same red-light delay as a private motorcycle, potentially impacting emergency response outcomes. While police officers provide a necessary human override capability during extreme events, manual control introduces further complications. A single officer cannot maintain awareness of traffic conditions on all approaches simultaneously, cannot coordinate with officers at adjacent junctions in real-time, and relies on heuristic judgment rather than systematic optimization. Improper manual management can inadvertently worsen congestion by creating imbalances that propagate through the network. According to the statistics for the year 2022 available with the Ministry of Transport, Highways, Ports, and Civil Aviation, the number of persons killed in road accidents involved 820 Motorcyclists, followed by 720 pedestrians. Overall, 2515 deaths took place in the year 2022 [2].

The aggregate effect of these inefficiencies manifests as substantial time delays for commuters during peak hours, contributing to lost economic productivity, increased fuel consumption and emissions from idling vehicles, and elevated frustration levels among road users. The problem is particularly acute during the evening peak period when workers return from Colombo to suburban areas, creating directional demand surges that fixed-time systems cannot accommodate.

7.4. Research Objectives

7.4.1. Main Objectives

The primary goal of this research is to create an intelligent traffic signal control system that uses a MARL framework based on centralized training with decentralized execution (CTDE) paradigm using value decomposition networks (VDN), enabling coordinated control across multiple junctions while maintaining per-junction autonomy during execution to dynamically optimize traffic flow depending on current conditions, while maintaining efficiency and safety. This includes integrating real-time detection inputs

from four computer vision-based components, namely, emergency vehicle detection, normal vehicle classification, pedestrian detection, and accident identification, into a unified MARL observation space, enabling the system to respond to dynamic multi-modal traffic conditions. A priority-aware reward function will be developed that simultaneously optimizes for emergency vehicle passage, pedestrian safety with weighted consideration for vulnerable groups, public transit efficiency, and overall network throughput, ensuring differentiated service levels aligned with societal priorities. Implementation of a lightweight inter-agent communication protocol enabling junctions to share pressure metrics and intended actions with neighbors, facilitating proactive coordination for emergency vehicle green waves and queue spillover prevention, remains a top objective.

7.4.2. Specific Objectives

- Develop an adaptive traffic signal control system using multi-agent reinforcement learning for multiple interconnected junctions
- Enable real-time traffic monitoring and response using integrated detection systems (vehicles, pedestrians, emergencies, accidents)
- Design agents to make decisions based on priority-aware traffic features, giving higher importance to emergency vehicles and vulnerable road users
- Achieve coordination between intersections through inter-agent communication to improve overall traffic flow
- Minimize vehicle waiting time, congestion, and unnecessary delays across the network
- Ensure priority handling of emergency vehicles by reducing stops and enabling faster passage
- Incorporate pedestrian-aware signal control to improve safety and reduce pedestrian waiting time
- Implement a scalable and decentralized execution framework where each junction operates independently after training
- Provide real-time visualization and control through a dashboard for monitoring and manual intervention

- Support multiple operational modes (MARL, fixed-time, manual) for flexibility and fallback
- Enable data-driven evaluation and comparison of different traffic control strategies
- Integrate with IoT-based physical traffic light systems for real-world applicability
- Ensure system robustness and reliability through fallback mechanisms and operator control

7.4.3. Business Objectives

The primary business objective is to optimize urban traffic flow and reduce congestion through an intelligent, adaptive traffic signal control system that improves travel efficiency, safety, and emergency response times.

- Improve overall traffic efficiency by reducing congestion and travel time across key urban junctions
- Enhance public safety by prioritizing emergency vehicles and improving pedestrian signal management
- Reduce fuel consumption and emissions, contributing to environmental sustainability
- Enable cost-effective traffic management by minimizing the need for manual traffic control (e.g., police intervention)
- Provide real-time monitoring and control for traffic authorities through an interactive dashboard
- Support data-driven decision making using analytics and performance comparisons across control strategies
- Increase road network productivity by optimizing vehicle throughput at intersections
- Ensure scalability so the system can be expanded to additional junctions or cities with minimal overhead
- Improve incident response time through automated detection and alert systems for accidents

- Facilitate smart city integration by connecting with IoT-based traffic infrastructure (e.g., Arduino-controlled signals)
- Enhance operational flexibility with multiple control modes (adaptive, fixed-time, manual)
- Reduce operational costs associated with inefficient traffic systems and delays
- Provide a future-ready platform that can integrate with advanced technologies like autonomous vehicles and smart transport systems

8. METHODOLOGY

8.1. Methodology

The methodology served as the foundation of this applied research, laying out the systematic processes, instruments, and strategies used to gather, process, and interpret data in accordance with the project's objectives. Particularly in projects involving

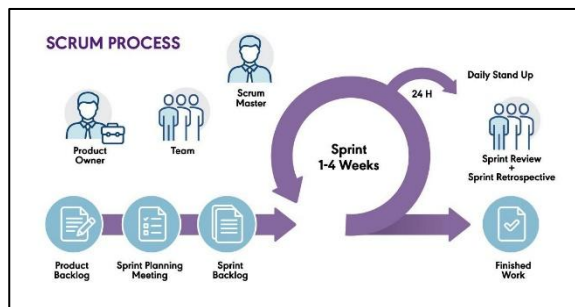


Figure 1: Agile Scrum Framework

computer vision, acoustic signal processing, and real-time traffic simulation, a clearly

defined methodology guaranteed that the development process was transparent, reproducible, and grounded in sound logic. To address the intricate and iterative requirements of building a multi-modal emergency vehicle detection system integrated with the SUMO environment, this study employed a hybrid methodology that combined a seven-stage System Development Life Cycle (SDLC) with Agile software development practices.

Agile Methodology for Research Development: The main project management framework for this study was Agile, a methodology originally created for software development but increasingly popular in multidisciplinary research due to its adaptability, iterative structure, and evolution driven by stakeholder feedback. Agile was the perfect choice for a dynamic project with changing traffic management requirements and evolving technical complexities, as it placed a strong emphasis on collaboration, modularity, and continuous testing. Core modules like YOLO-based visual detection, OCR label extraction, acoustic siren recognition, and hue-based light bar detection were able to be seamlessly integrated into the SUMO simulator thanks to the Agile model.

In particular, the Scrum version of Agile was used, in which the work was broken down into smaller units called sprints, which were usually two to three weeks long. A major system component (such as multi-angle visual processing, acoustic model training, or TraCI signal pre-emption logic) was the focus of each sprint. Continuous improvement and prompt issue resolution were made possible by regular sprint reviews, daily stand-up meetings, and retrospectives. The Scrum cycle used in this study is depicted in Figure 5, which shows how tasks moved through iterative feedback loops from planning to delivery.

Seven-Stage Development Framework: Based on the ideas of Agile methodology, the research process developed over the course of a structured seven-stage process. Requirements collection, system design, deployment, testing, maintenance, and continuous improvement were all phases in this process. The results and input from the previous stage were used to inform each phase, resulting in a dynamic and adaptable development loop. The team was able to address issues as they arose, improve technical elements, and adjust to changing system needs thanks to this iterative structure. Agile's focus on adaptability was especially helpful in handling the project's multidisciplinary elements, which integrated cutting-edge computer vision and acoustic machine learning methods with real-time traffic simulation constraints. The development cycle encouraged methodical advancement while providing room for creativity, user input, and iterative improvement. Figure 6 shows the seven-stage development life cycle, and the rest of the methodology section will explain each of the phases of the life cycle.

Iterative Development and Collaboration: Cross-functional cooperation and incremental development cycles were key factors in the multi-modal detection system's success. To evaluate module performance and adjust to evolving needs, the interdisciplinary team—which included software developers, computer vision engineers, and traffic simulation experts—convened biweekly review sessions. Agile collaboration tools like Jira and GitHub Projects made it easier to track issues, documentation, and code changes in real time. This iterative approach proved particularly helpful in

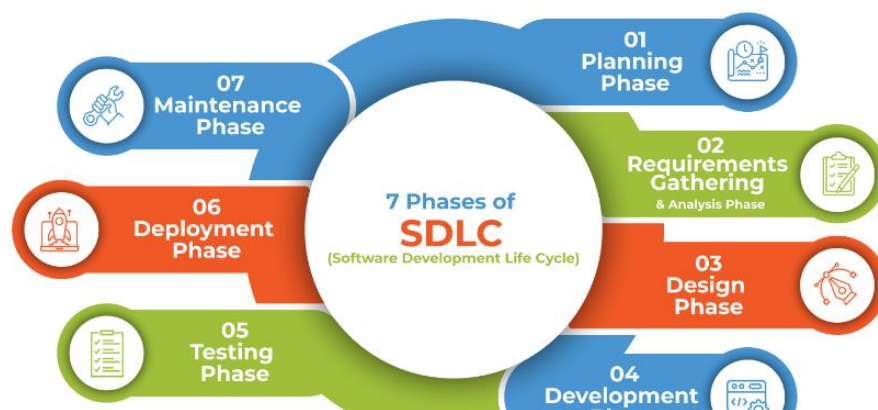


Figure 2 : Seven Stage of Software Development Life Cycle

improving YOLO detection accuracy under varying lighting conditions, filtering urban noise from the acoustic siren detection, and fine-tuning the TraCI connection with the SUMO environment.

Flexibility and Responsiveness: Agile methodology's adaptability and responsiveness were essential in overcoming the ever-changing obstacles that arose during the detection system's development. Unexpected problems like latency in real-time video processing, false positives in HSV light bar detection, and synchronization delays between the detection modules and the SUMO simulator surfaced as the project progressed.

Because Agile is iterative, the team was able to quickly adjust, optimizing OpenCV pipelines, refining acoustic ML models, and improving the logic for traffic signal pre-emption without causing the project's overall timeline to be delayed. This ability to react quickly to changing technical and simulation requirements made sure the system continued to meet performance and latency objectives. Eventually, while preserving development continuity and responsiveness to testing feedback, Agile principles made it possible to successfully integrate sophisticated features like multi-modal sensor fusion, dynamic traffic signal pre-emption, route prediction, and dashboard visualization.

8.1.1. Feasibility Study / Planning

The feasibility study examines the viability of designing and implementing a multi-modal emergency vehicle detection system integrated with the SUMO traffic simulator. This system encompasses features such as real-time visual detection from frontal and top-view cameras, OCR-based label extraction, acoustic siren recognition, and hue-based light bar detection to dynamically trigger traffic signal pre-emption. The present section provides a comprehensive analysis of the technical, operational, economic, and legal feasibility of the proposed solution.

1. Technical Feasibility:

- **Data Availability and Field Collection:** Following the initial project proposal and subsequent panel suggestions, the data collection strategy was significantly enhanced to include OCR-based top-angle detection, specifically designed to

identify emergency vehicles caught in traffic blockages. To ensure high-fidelity, real-world accuracy, extensive primary data was gathered through targeted field visits. Fire engines were physically inspected at the Vavuniya Municipal Council, ambulances were examined at the Vavuniya Hospital and at SLIIT (in collaboration with the 1990 emergency service), and police vehicles were inspected at the Cheddikulam Police Station. This fieldwork allowed for the capture of distinct vehicle types, structural labels, visual profiles under varying lighting, and crucially, the specific acoustic differences between ambulance and fire engine sirens.

- **Hardware and Software Requirements:** The working prototype was developed and executed on a standard laptop capable of handling concurrent video and audio streams. Real-time visual and acoustic inputs were captured utilizing standard webcams, with the incoming data stored in a database and continuously retrieved for real-time processing. The software stack comprised YOLO, OpenCV, and Tesseract for visual and OCR processing, alongside YAMNet for acoustic siren detection. To physically demonstrate the system's traffic pre-emption capabilities, an Arduino board was integrated to dynamically visualize and control the traffic signal changes triggered by the detection algorithms, successfully bridging the gap between software processing and hardware actuation.
 - **Model Development:** The system utilizes a multi-modal fusion pipeline. YOLO processes the visual feeds, while HSV color segmentation and Haar Cascade classifiers are deployed specifically for light bar recognition. Simultaneously, a machine learning acoustic model processes audio feeds to isolate siren frequencies. The OCR module was specifically trained on the top-view angles captured during fieldwork to extract vehicle labels when frontal views are obstructed in a traffic jam.
2. **Economic Feasibility:** This study considered both development and operational costs. The total cost of the project was estimated at LKR [Insert your actual total cost], as detailed in Table 1.
- **Resource Allocation:** The development team comprised researchers possessing specialized knowledge in computer vision, acoustic machine learning, traffic simulation (SUMO), and legal compliance. Responsibilities were allocated based

on individual strengths, with specific members assigned to conduct the primary field research.

- Return on Investment (ROI): The platform provides considerable return on investment by significantly reducing emergency response times (ERT), minimizing fuel wastage caused by traffic idling, and potentially saving lives and property by ensuring unobstructed paths for first responders.
3. Legal and Ethical Feasibility:
- Legal Compliance (Sri Lankan Context): A critical aspect of this study's feasibility involved validating the system's detection criteria against national laws to prevent false pre-emption. Consultation with a legal professional confirmed the parameters outlined in the Motor Traffic (Amendment) ACT No. 8 of 2009, Section 67. The system's acoustic and hue-based detection algorithms were strictly engineered to recognize only legally defined emergency signals. Specifically, the system was programmed to trigger pre-emption only when detecting:
 - *Acoustic limits*: Multi-tone horns or harsh, shrill alarms, which the Act states may only be used by the fire brigade, police/armed services, or an ambulance responding to an emergency call.
 - *Visual limits*: Specific light combinations mandated by the Act, including:
 - Red light for the Fire Brigade.
 - Red and blue lights for Police vehicles.
 - Red light for Ambulances.
 - Blinking blue or amber light for other Commissioner-General approved emergency vehicles.
 - Data Privacy and Ethics: The system utilizes simulated and public traffic camera feeds. The algorithms were designed strictly for vehicle and siren detection, with no capability to retain personally identifiable information (PII) of drivers or pedestrians.
4. Operational Feasibility:
- Multi-Modal Processing: The system proved highly capable of simultaneously analyzing multiple data streams. The addition of the panel-suggested top-view OCR module ensured that even if a vehicle is boxed in by civilian cars

(obstructing the frontal view), the system can still read the "AMBULANCE" or "FIRE" labels from above.

- **System Deployment and Integration:** The detection engine was successfully coupled with the SUMO traffic simulator via TraCI. For the purposes of this research, a centralized processing architecture was chosen to handle the YOLO, acoustic, and OCR pipelines before sending the pre-emption command to TraCI. The dashboard visualization was deployed to effectively relay this real-time data to end-users and relevant authorities.
5. **Time/Schedule Feasibility:**
- **Project Timeline:** A comprehensive timeline was created, including essential milestones such as the initial proposal, the panel review (which prompted the addition of the top-view OCR), the field data collection phase in Vavuniya/Cheddikulam, legal consultation, model training, SUMO integration, and final testing. The iterative Agile methodology provided room for adaptability when integrating the panel's feedback and the legally mandated light/sound constraints. Figure 7 illustrates the Schedule Management plan.

Figure 7: Gantt Chart (Schedule Management)

6. **Social and Cultural Feasibility:**
- **Acceptance and Impact:** The tool addresses a critical urban pain point , traffic congestion delaying emergency services. By automating the clearing of intersections, the system reduces stress for both commuters and first responders.
 - **Bias, Fairness, and Public Trust:** Because the system's triggering mechanism is legally bounded by the Motor Traffic Act No. 8 of 2009, it eliminates the risk of "favoritism" or false triggering for unauthorized vehicles using illegal multi-tone horns or inappropriate flashing lights. This strict adherence to the legal definitions of red, blue, and amber emergency lights ensures public trust and prevents unnecessary traffic disruptions.

Risk Management Plan

To anticipate and mitigate problems with team availability, supervisor input, and stakeholder involvement, a comprehensive risk management plan was created. The risk management strategy is displayed in Table 2.

Communication Management Plan

The Communications Management Plan is designed to ensure that all team members, as well as the supervisor and co-supervisor, receive the necessary information to effectively perform their roles throughout the project. Successful project execution relies on strategic planning and effective communication. This plan outlines how to communicate with various stakeholders efficiently, detailing the audience, content, format, frequency, and expected outcomes of communications. It also specifies each stakeholder's role, task assignment, and communication strategy based on their influence, interests, and expectations.

Communication Objectives Proactive communication is essential for project success. Communication should be:

- Adequate: Providing information in the right format and content.
- Specific: Tailored to the targeted audience.
- Sufficient: Including all necessary details.
- Concise: Clear and to the point, avoiding unnecessary repetition.
- Timely: Addressing relevant points at appropriate times.

Communication Media The following media will be utilized for communication throughout the project:

1. Microsoft Teams: For formal updates, documentation, and detailed communications.
2. Documents (MS Word and PowerPoint): For reports, presentations, and detailed project documentation.
3. Phone Calls: For quick discussions and urgent matters.

4. Meetings: Conducted via meeting rooms, conference phones, or Microsoft Teams for in-depth discussions, planning, and progress reviews.
5. Chats (WhatsApp): For informal, quick exchanges and real-time updates.

8.1.2. Requirement Gathering & Analysis

The requirement analysis phase relied heavily on stakeholder consultation and real-world domain analysis to extract both functional and non-functional system requirements. A pivotal shift in the project's scope occurred during the panel evaluation, which mandated a new functional requirement for top-angle OCR to handle traffic congestion scenarios. To translate this requirement into actionable, ground-truth data specifications, direct engagement with emergency service providers was necessary. Site inspections were conducted at the Vavuniya Municipal Council, Vavuniya Hospital, SLIIT (in collaboration with 1990), and the Cheddikulam Police Station. This empirical approach allowed the research team to extract precise technical requirements for the ML pipelines, including exact text placement for OCR parsing, non-functional requirements for handling dynamic lighting, and specific acoustic frequency ranges required to accurately classify different emergency vehicle sirens.

Data Collection

To achieve robust detection under varied conditions, a multi-source dataset acquisition strategy will be used:

- **Public Web Scraping:** A foundational visual dataset was established by sourcing a diverse array of emergency vehicle images (ambulances, police cars, and fire trucks) from public web engines. These images were specifically selected to represent various lighting, weather, and traffic conditions, and underwent strict filtering for quality, relevance, and resolution prior to preprocessing.
- **Open-Source Repositories:** To supplement the custom data, several pre-labeled datasets from Kaggle were utilized. These included high-resolution emergency

vehicle image datasets with bounding box annotations, urban traffic object detection datasets (to provide a balanced mix of emergency and non-emergency vehicles), vehicle indicator light datasets (for route prediction), and the UrbanSound8K dataset (to isolate baseline siren audio for acoustic training).

- Primary Field Recording: To fine-tune the models for real-world application, authentic data was captured by deploying mobile and static cameras at local intersections to record actual emergency vehicle footage. Concurrently, directional microphones were used to record siren audio across varying levels of ambient urban noise, ensuring the acoustic training data accurately reflected real-world conditions.

Project Requirements

The project requirements define the expected functionality and operational qualities of the Emergency Vehicle Detection and Triggering Signal in SUMO Simulator component. These requirements ensure that the system meets both performance goals and real-world applicability.

Functional requirements delineate the precise features and functionalities that a system must incorporate to achieve its designated traffic management and emergency response goals. These requirements served as a fundamental basis for the design, development, and validation processes of the detection system.

- Stakeholder Input Approach: The requirement identification process extended beyond theoretical research, relying heavily on primary stakeholder engagement and expert consultation:

- Crucial system enhancements were derived directly from the initial project panel review, which identified the specific need for top-angle OCR detection to handle vehicles trapped in traffic blockages.

- Operational and technical requirements were gathered through targeted field visits with first responders. Personnel at the Vavuniya Municipal Council, Vavuniya Hospital, the 1990 ambulance service, and the Cheddikulam Police Station provided insights into actual vehicle deployment, structural labeling, and the practical acoustic variations between different emergency sirens.

➤ Additionally, ongoing consultations were held with a legal professional who assisted the development team in comprehending the strict legal boundaries of emergency signal usage, ensuring the system's triggering mechanisms were fully compliant with national regulations.

Figure 8 illustrates the details of the field visits and stakeholder consultations conducted to gather technical specifications, vehicle profiles, and legal constraints from emergency service personnel.

Figure 8: Field Visit and Stakeholder Consultation Details

In addition to the collection of functional requirements, this phase encompassed the manual compilation of a highly specific, multi-modal dataset designed for the training and validation of the detection models. The ground-truth data was exclusively sourced from direct physical observations and legal statutes during the field visits, which included:

- Visual captures of distinct vehicle types, structural labels, and top-angle profiles (for YOLO and OCR training)
- Acoustic recordings of specific siren frequencies under varying noise conditions (for acoustic model training)
- Visual documentation of specific light bar configurations (for HSV segmentation training)
- Legal text excerpts from the Motor Traffic (Amendment) ACT No. 8 of 2009, Sec 67 (for rule-based system validation)

All resources were verified on-site and against legal documentation to ensure complete adherence to real-world emergency vehicle configurations and Sri Lankan traffic laws, thereby guaranteeing that the system's detection criteria maintain operational accuracy and legal compliance. Figures 9, 10, and 11 illustrate examples of the captured real-world vehicle profiles, structural labels, and the specific light bar configurations documented during the field visits.

8.1.3. Functional Requirements

Identification of Key Functionalities:

- Visual Detection -Accurately detect emergency vehicles in real-time using a YOLO-based model and OpenCV video processing.
- Acoustic Detection -Identify ambulance sirens using DSP techniques and machine learning models even in noisy urban environments.
- Hue-Based Light Bar Detection -Detect emergency vehicle light bars using HSV colour segmentation and Haar Cascade classifiers.
- Multi-Modal Fusion -Combine visual, acoustic, and hue detection outputs to increase detection accuracy and reduce false positives.
- Indicator Detection -Detect turn indicators of emergency vehicles to predict movement routes.
- SUMO Integration -Communicate with SUMO via TraCI API to pre-emptively adjust traffic light phases for emergency vehicle passage.
- Route Sharing -Transfer predicted routes to neighbouring simulated intersections for coordinated signal adjustments.
- Dashboard Visualization -Display real-time detection events, simulation updates, and traffic light statuses on a web-based dashboard.
- Data Logging -Store detection events, simulation changes, and performance metrics for later analysis.
- Authority Notification -Send alerts to relevant traffic management authorities when emergency vehicles are detected.

Non-Functional Requirements

The system shall:

- Performance -Operate with a detection latency of less than 500 ms per frame for visual and acoustic detection combined.
- Accuracy -Achieve an overall multi-modal detection accuracy of at least 95% in simulation testing.
- Scalability -Support integration into larger Intelligent Transportation Systems (ITS) with minimal modifications.

- Reliability -Maintain consistent detection performance under varying lighting, weather, and noise conditions.
- Security -Protect stored and transmitted data through encryption and secure communication channels.
- Maintainability -Ensure modular code structure with proper documentation for easy updates and debugging.
- Usability -Provide an intuitive dashboard interface that can be operated without advanced technical knowledge.
- Compatibility -Be deployable on both local GPU-enabled systems and cloud-based environments.
- Resource Efficiency -Optimize computational resource usage to run on mid-range hardware without significant performance drops.

8.1.4. Designing

8.1.5. Implementation

8.1.6. Testing

8.1.7. Deployment & Maintenance

8.2. Development Methodology

8.2.1. Visual Detection Module

Description: Implement a YOLO-based real-time detection module integrated with OpenCV to identify emergency vehicles from live or simulated traffic camera feeds. The model will be fine-tuned with a dataset containing diverse lighting, weather, and traffic conditions.

Technologies & Tools:

- YOLOv8 / YOLOv5 (PyTorch) -Object detection framework for real-time identification.
- OpenCV -For video frame extraction, preprocessing, and drawing detection bounding boxes.
- Python -Core programming language for detection and integration.
- CUDA -For GPU acceleration during detection tasks.

8.2.2. Audio Detection Module

Description: Develop a siren detection system using Audio detection and machine learning to recognize ambulance sirens in noisy urban soundscapes. Audio input will be processed to extract features such as MFCC for model training.

Technologies & Tools:

- Librosa -For audio signal processing and feature extraction.
- TensorFlow / Keras -For building and training the siren classification model (CNN/LSTM).
- Scikit-learn -For preprocessing and baseline ML models.
- PyAudio -For capturing audio streams in real-time.

8.2.3. Hue-Based Light Bar Detection Module

Description: Implement HSV colour segmentation to detect characteristic emergency vehicle light bars, combined with Haar Cascade classifiers for improved detection reliability under varying conditions. Figure 1 shows the sound detection of siren using NN to detect it and give the single output.

Technologies & Tools:

- OpenCV -For HSV colour space segmentation and image filtering.
- Haar Cascade Classifiers -Pre-trained or custom-trained for light bar detection.
- Python -For scripting and integration with other modules.

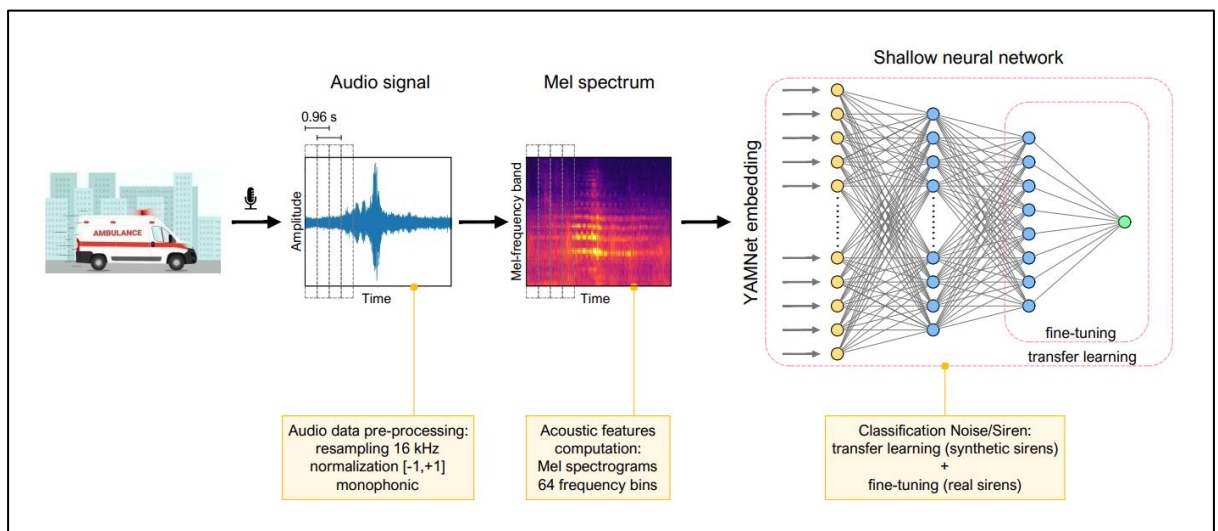


Figure 3 : Audio Siren Detection Source [14]

8.2.4. SUMO and TraCI Integration

Description: Connect the detection system to the SUMO traffic simulation environment using the TraCI API to trigger and adapt traffic light phases in real-time upon detecting an approaching emergency vehicle.

Technologies & Tools:

- SUMO (Simulation of Urban Mobility) -Traffic simulation software.

- TraCI (Traffic Control Interface) -Python API for controlling simulation events.
- NetEdit -SUMO network editor for creating and configuring intersections.

Figure 2 is the virtual environment where the traffic-based programs can be done it is more flexible and portable to use.



Figure 4 : SUMO Stimulator , Source : <https://networksimulationtools.com/sumo-simulator/>

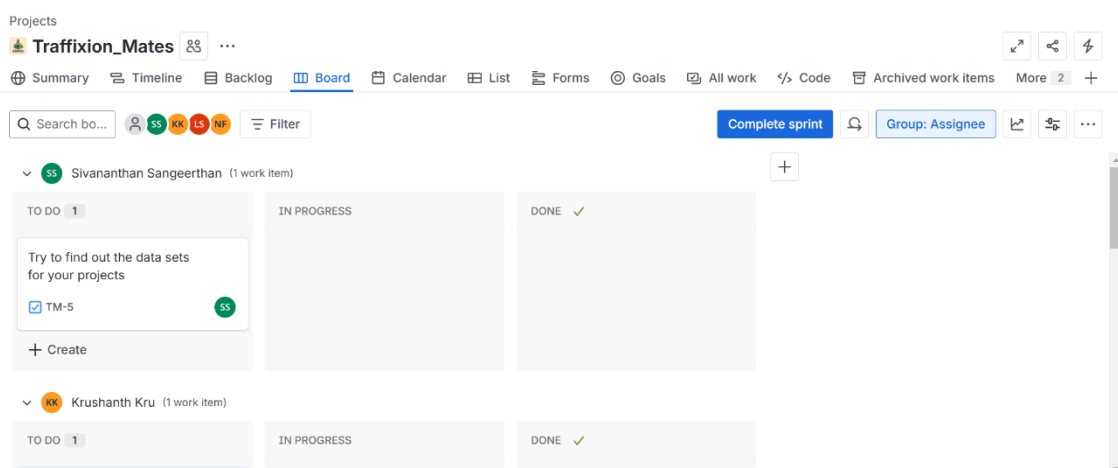


Figure 5 : Jira management tool

8.3. System Analysis

8.3.1. Software Solution Approach

The proposed system will be developed as a modular, multi-modal detection and simulation platform that integrates visual, acoustic, and hue-based detection techniques with the SUMO traffic simulation to enable real-time emergency vehicle prioritization.

Data Collection -Gather emergency vehicle image datasets (Google, Kaggle, real-world) and siren audio datasets (UrbanSound8K, Kaggle, recordings).

Pre-processing -Augment and normalize images.

Model Training -Use YOLOv8/YOLOv5 for visual detection, HSV + Haar Cascade for light bar detection.

Multi-Modal Fusion -Combine visual, audio, and hue-based outputs via decision fusion to improve accuracy.

SUMO Integration -Control traffic signals dynamically with TraCI when emergency vehicles are detected.

Route Prediction -Detect turn signals to predict paths and share with nearby intersections.

Dashboard -Real-time monitoring via React.js frontend and Flask/FastAPI backend, with charts/maps.

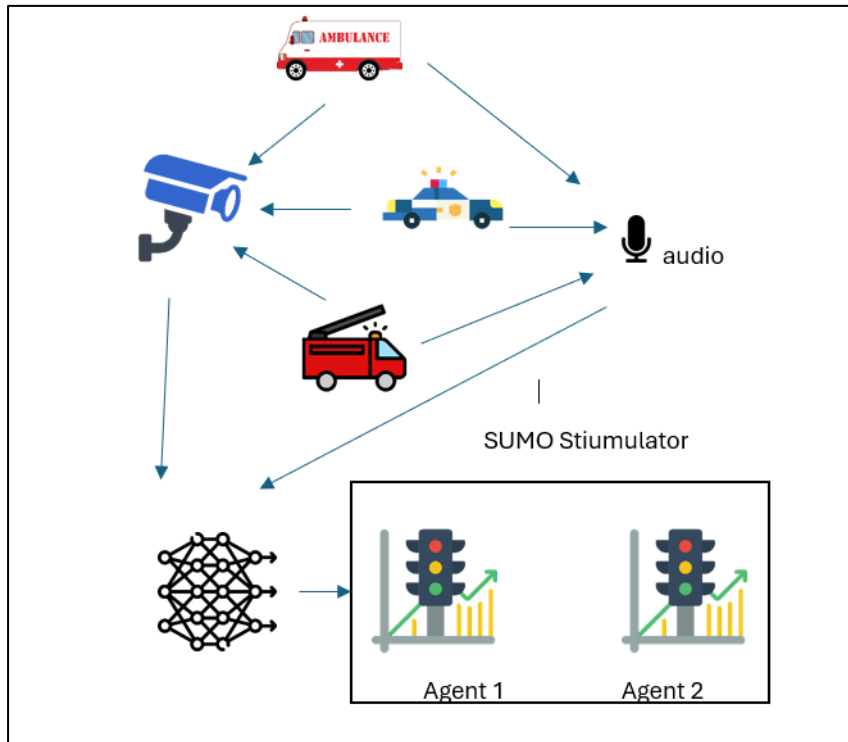


Figure 6 : System Diagram

8.3.2. Tools and Technologies

The proposed system will be developed using a combination of software frameworks, programming languages, simulation tools, and development platforms to ensure robustness, scalability, and real-time performance.

Programming Languages

- Python 3.x -The core development language for computer vision, audio processing, machine learning, and SUMO integration due to its extensive library ecosystem.
- JavaScript (React.js) -For building the real-time interactive dashboard frontend

Frameworks and Libraries

- YOLOv8 / YOLOv5 (PyTorch) -State-of-the-art deep learning frameworks for real-time visual emergency vehicle detection.
- OpenCV -Computer vision library for video frame processing, HSV colour segmentation, and Haar Cascade implementation.

- TensorFlow / Keras -Machine learning frameworks for training CNN/LSTM models for siren sound recognition.
- Librosa -Python library for audio feature extraction such as MFCC, used in acoustic detection.
- Scikit-learn -For data preprocessing, model evaluation, and integration of classical ML methods.

Traffic Simulation and Integration Tools

- SUMO (Simulation of Urban Mobility) -Microscopic, open-source traffic simulator for testing signal pre-emption strategies in urban environments.
- TraCI API -Traffic Control Interface for real-time control of SUMO simulations through Python scripts.
- NetEdit -SUMO's network editor for building and modifying traffic network layouts.

Data Collection and Annotation Tools

- Google Images / Kaggle Datasets -Sources for diverse emergency vehicle, siren, and traffic datasets.
- LabelImg / Roboflow -Annotation tools for creating bounding box labels for training YOLO models

Backend Development and API Tools

- Flask / FastAPI -Python-based web frameworks for creating RESTful APIs to serve detection and simulation data to the dashboard.

Frontend Development and Visualization Tools

- React.js -For creating a responsive, real-time web dashboard for visualizing detections and simulation status.

Development and Project Management Tools

- VS Code -Main integrated development environment for coding and debugging.

- Google Colab -For GPU-enabled training of detection and audio models.
- Jira -For Agile sprint planning, backlog management, and task tracking.
- Git & GitHub -For version control and collaborative development.

Testing Tools

- Pytest -For automated unit and integration testing of Python modules.
- Postman -For testing and validating API endpoints.

Hardware Requirements

- GPU-enabled Workstation / Cloud GPU -For training deep learning models efficiently.
- CCTV / IP Cameras -For real-world visual data collection and live feed integration.
- Directional Microphones -For real-time siren sound recording in urban environments.

8.4. Testing

Testing is a critical phase in the development of the Emergency Vehicle Detection and Triggering Signal in SUMO Simulator component to ensure system reliability, accuracy, and real-time performance. Multiple testing approaches will be applied to validate individual modules, integrated functionality, and overall simulation outcomes.

8.4.1. Unit Testing

Purpose: To validate that each independent module of the system works as intended in isolation before integration.

Scope:

- Visual Detection Module: Verify YOLO model correctly detects emergency vehicles in test images and videos with expected accuracy.
- Acoustic Detection Module: Ensure siren recognition model correctly classifies audio clips as siren or non-siren.
- Hue-Based Detection Module: Test HSV segmentation and Haar Cascade to ensure accurate light bar detection under different lighting conditions.
- Indicator Detection Module: Confirm detection of turn signals and extraction of route prediction data.

8.4.2. Integration Testing

Purpose: To verify that all modules work together correctly when combined into the complete system.

Scope:

- Integration of visual, acoustic, and hue-based detection outputs into the multi-modal fusion logic.
- Communication between detection modules and the SUMO simulation via TraCI API.
- Route prediction data transfer to neighbouring simulated intersections.
- Real-time update flow from backend to dashboard using REST API.

8.4.3. Performance Testing

Purpose: To assess the system's responsiveness, accuracy, and computational efficiency.

Metrics:

- Detection Latency: Measure the time taken from receiving visual/audio input to generating detection results.
- Detection Accuracy: Evaluate multi-modal fusion accuracy against a ground-truth dataset.
- System Throughput: Test number of frames processed per second (FPS) under GPU and CPU configurations.
- Resource Utilization: Monitor CPU, GPU, and memory usage during peak operation.

9. RESULTS & DISCUSSION

10.FUTURE SCOPE

The Emergency Vehicle Detection and Triggering Signal in SUMO Simulator component has strong potential for commercialization in both public sector and private sector markets. The system addresses a critical urban challenge reducing emergency response times making it valuable for deployment in Intelligent Transportation Systems (ITS) worldwide.

Potential Markets

- Government Traffic Management Authorities
 - City councils and transportation departments can integrate the system into existing traffic control centres to optimize emergency vehicle movement.
- Emergency Services
 - Ambulance, fire, and police departments can deploy the solution to ensure faster and safer passage through congested urban intersections.
- Smart City Solution Providers
 - Technology vendors offering IoT-based traffic management systems can incorporate the multi-modal detection module as a premium feature.

Future Commercial Potential

The technology can be extended beyond emergency vehicles to support:

- Public Transport Priority: Buses and trams receiving traffic signal priority during peak hours.

- **Autonomous Vehicle Integration:** Providing signal priority for driverless emergency and service vehicles.

11.CONCLUSION

The proposed Emergency Vehicle Detection and Triggering Signal in SUMO Simulator component is a multi-modal intelligent traffic management solution designed to prioritize emergency vehicles in urban environments. By integrating YOLO-based visual detection and machine learning-based siren recognition, and HSV Cascade light bar detection, the system achieves high detection accuracy under varying conditions. Real-time integration with SUMO and TraCI enables dynamic traffic signal pre-emption, reducing response times and enhancing road safety.

The project follows an Agile development methodology, ensuring iterative improvement through testing phases including unit, integration, simulation, and performance testing. Expected outcomes include reduced travel times for emergency vehicles, improved intersection safety, and a scalable foundation for real-world deployment. With strong commercialization potential in government traffic control, emergency services, and smart city solutions, the system offers a cost-effective, high-impact contribution to modern urban mobility management.

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